SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1526

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I Latisha. S (192524071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Latisha.

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :

CO2 - AT1

1.

* A hypervisor is software that creates and manages virtual Machines (VMs), allowing multiple operating systems to run on a single physical server

Feature.	Type-1 Hypervisor	Type-2 Hypervisor.
Working	Runs directly on hardware	Runs on a host operating system.
Performance	High performance	comparatively lower performance
Security	Less dependent on a general purpose host OS	Depends on host OS security.
Management	Suitable for centralized management	Suitable for desktop management.
Examples	VMware ESXi, Xen	Oracle VirtualBox, VMware workstation

Analysis :

1. Performance :

Type-1 hypervisor provide direct access to hardware and are suitable for high performance applications. Type-2 hypervisors operate through a host OS, which adds overhead.

2. Security:

Type-1 hypervisors provide strong VM isolation when properly configured. Type-2 hypervisors also isolate VMs, but vulnerabilities in the host OS may affect them.

3. Manageability:

Type-1 hypervisors support centralized resource allocation, VM migration and monitoring. Type-2 hypervisors are easier to use for testing and development

* Type-1 hypervisors are the better choice for mission-critical cloud applications because they offer high performance, effective resource management and suitable isolation capabilities.

* Type-1 hypervisors are preferred for production cloud data centers, while Type-2 hypervisors are mainly used for testing and development.

2.

Virtualization allows multiple Virtual Machines to run on the same physical server while maintaining separate operating environments. It helps protect sensitive data belonging to different departments.

How Virtualization supports isolation:

1. Virtual Machine isolation:

Finance, healthcare and student portal applications can run in separate VMs with independent operating systems and resources.

2. Network isolation:

Virtual networks and firewalls restrict unauthorized communication between departments.

3. Storage isolation:

Separate virtual disks and access permissions protect confidential data.

4. Access control:

Role-based access control ensures that users access only the resources they are authorized to use.

Two major attack surfaces:

1. Hypervisor attacks:

The hypervisor controls all virtual machines. If an attacker exploits a hypervisor vulnerability, they may attempt to escape from one VM and access other VMs.
So, we apply patches for security, restrict administrator access and monitor hypervisor activity.

2. Virtual network attacks

Attackers may exploit misconfigured virtual switches, weak network rules or exposed management interfaces. They may attempt unauthorized communication between finance, healthcare and student systems.

3. Given Values

Parameters	Observed value
CPU utilization	92%
Memory utilization	88%
Storage I/O latency	35ms
Average VM boot time	170 seconds.

Analysis of bottlenecks

1. CPU utilization - 92%.

CPU utilization is very high, meaning that the host has limited processing capacity available for additional workloads.

- Possible cause: Too many active VMs or CPU-intensive applications.
- Corrective action: Migrate some VMs to another host, balance workloads, reduce unnecessary process or add CPU capacity.

2. Memory utilization – 88%.

Most of the host's memory is already used and is in use. This can cause memory contention and swapping.

- Possible cause: Insufficient RAM or excessive memory allocation to VMs.
- Corrective action: Add RAM, optimize VM memory allocation and reduce unnecessary workloads.

3. Storage I/O latency – 35ms.

Storage latency measures the time taken to complete storage operations. A latency of 35ms may be high for latency-sensitive applications, depending on the storage system and workload.

- Possible cause: slow disks, heavy simultaneous read/write operations or storage contention.
- Corrective action: use faster SSD or NVMe storage, distribute I/O-intensive VMs and optimize storage performance.

4. Average VM boot time – 170 seconds.

VMs are taking almost three minutes to start, which may delay service availability.

- Possible cause: CPU and memory contention, slow storage or lengthy operating system startup processes.
- Corrective action: optimize startup services, investigate storage delays and reduce resource contention.

*. The host is experiencing high CPU and memory utilization, along with potentially problematic storage latency. These conditions may contribute to slow VM boot times. Resource optimization, workload migration and storage improvements should be prioritized.

4. Introduction:

* A software defined datacenter (SDDC) is a datacenter in which compute, storage and networking resources are virtualized and managed using software.

* In traditional datacenter, many tasks depend on physical hardware configuration and manual administration. SDDC automates these tasks, making infrastructure more flexible.

Traditional Datacenter VS SDDC.

Feature	Traditional Datacenter	SDDC..
Compute	Physical servers are configured manually	VMs are created and managed through software
Storage	Mainly managed through physical storage system	Virtual storage can be allocated dynamically.
Networking.	Physical network configuration is common	Virtual networks and software-defined policies are used
Resource allocation	Often slower and manual	Faster and more automated.
Scalability	May require hardware installation and configuration	Resources can be allocated dynamically, subject to available capacity

Role of Virtualization in SDDC.

1. compute Virtualization:
   → Creates multiple VMs on physical servers.
   → Allocates CPU and memory as needed
   → Benefit : faster deployment and better resource utilization.

2. Storage Virtualization:
   → combines storage into logical pools
   → Allocates storage as required.
   → Benefits : Easy storage management and expansion.

3. Network Virtualization:
   → Creates virtual networks, switches and routers.
   → supports network security and automation
   → Benefit : Faster networking and better isolation.

How SDDC improves Agility.

- Rapid deployment
- Automation
- Scalability
- Cost efficiency
- Disaster recovery

*. SDDC improves agility through virtualization, automation and flexible resource management, making it faster and easier to manage than a traditional datacenter.

5. Introduction

* Multi-cloud computing means using services from multiple cloud providers. Cloud federation allows these separate environments to cooperate and share resources under agreed policies
* Identity mismatches occur when cloud providers use different user identifies, authentication methods, roles or permission formats.

Analysis of problem

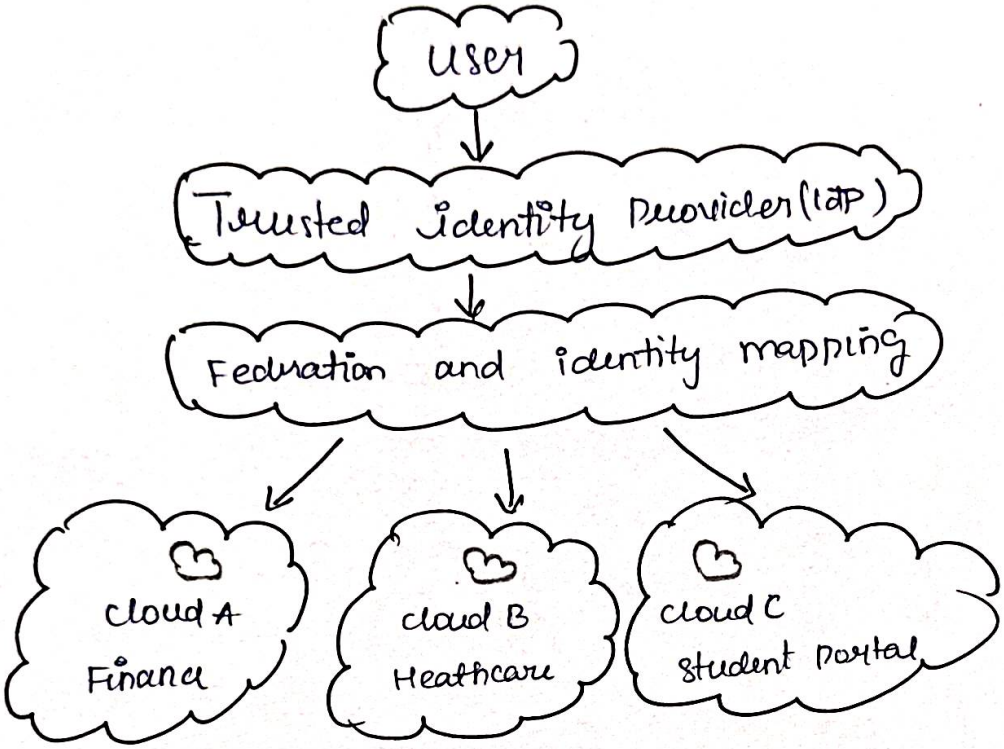
1. Different identity formats: One provider may identify a user using an email address, while another uses a different user ID.
2. Authentication differences: Providers may support different authentication protocols or configurations.
3. Role mismatches: A role such as "administrator" in one cloud may have different permissions in another.
4. Access failure: Users may be unable to access legitimate resources because their identities cannot be mapped correctly.
5. Security Risks: Incorrect identity mapping may grant unauthorized access to sensitive data.

Proposed Secure Identity and privacy design.

* Federated identity Management
* Identity mapping
* Role based access control (RBAC)
* Single sign-on (SSO).

* Multi - Factor authentication (MFA).
* Privacy protection.

Proposed Architecture



*. Identity mismatches in multi-cloud federation can cause access failures and security vulnerabilities. A trusted identity provider, standardized federation protocols, consistent role mapping, MFA and privacy protection create a secure and reliable multi-cloud environment.